

Undergraduate Science Laboratory (BS 192)

Chemistry Section

Name:

Roll Number:

Date: 22.02.2024

Signature:

1. How many ml of hydrogen (under normal pressure and temperature) do you expect from complete splitting of 1 ml of H_2O . (point: 1)

2. Write down the balanced reactions happening in cathode and anode during the electrolytic water splitting process. Write down the minimum energy requirement associated with these processes. (points: 1+1, 0.5+0.5)

3. What will happen if you use Cu wire as electrodes instead of graphite rods during the water electrolysis experiment under the same experimental conditions? (2 points)

4. Write down the Beer-Lambert law (define each term in the equation)? (point: 1)

5. Write down the possible reactions involved during the photochemical degradation of Methylene blue in presence of the iron oxide nanoparticles (points: 2)

6. Imagine that the LED light used in the photochemical degradation of Methylene blue (MB) has wavelengths between 450 nm and 700 nm. Will photochemical degradation of MB happen if you use TiO_2 (band gap of 3.2 eV) as the photocatalyst? Explain your result (points: 2)

7. Write down the chemical reaction between I_2 and ammonium oxalate in presence of light. (point: 1)

8. For a first order photochemical chemical reaction $A \rightarrow B$, plot how $[B]$ changes with time. Take the initial concentration of B is zero. Now imagine that you can change the intensity of the light. How will $[B]$ vs time plot will be affected if you increase the LED power? (point: 1+ 2)

9. What is the difference between systematic error and random error? How do you distinguish these two errors through practical experiments? (point: 1+1)
10. If you need to make 40 mL, 20 mM (mM = millimolar) solution of NaCl in water starting from a 5 M NaCl solution, how you will do that? (Point: 1)
11. Write down the balance equation of the reaction between L-alanine and Ninhydrin. What is the common name of the main product of this reaction? (point: 2+1)
12. Which amino acid does not give dark purple/blue colour with Ninhydrin and why? (Point: 1+1)
13. Which other chemical one can use for fingerprint test as a substitute of Ninhydrin? (point: 1)
14. Write down the chemical reaction of Nylon 6 and Nylon 6,6 synthesis (Point: 1+1)
15. In which regions do you expect IR peaks for C-O, C-H and N-H and how they will differ from the starting materials? (point: 1+1)
16. Why industry use basic condition for Nylon 6,6 synthesis and what will happen if one use neutral reaction condition? (point: 1)
17. What are the applications of Nylone 6 and Nylon 6,6? (point: 1)